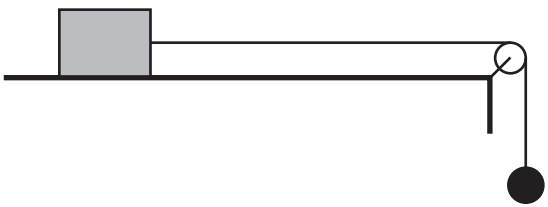
- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

11 A block of mass 3 kg is at rest on a smooth horizontal table. It is attached to a light inextensible string which passes over a smooth pulley. This part of the string is horizontal. A sphere of mass 1.2 kg is attached to the other end of the string. The sphere hangs with this part of the string vertical as shown in the diagram. A horizontal force of magnitude F N is applied to the block to prevent motion.



(a) Complete the copy of the diagram in the Printed Answer Booklet to show all the forces acting on the block and the sphere. **[2]**

(b) Find the value of F. **[2]**

The force F N is removed, and the system begins to move.

(c) The equation of motion of the block is $T = 3a$, where T N is the tension in the string and $a\,\mathrm{m\,s^{-2}}$ is the acceleration of the block.

Write down the equation of motion of the sphere. **[1]**

(d) Find the value of T. **[2]**

